ML and DL Powered Behavioral Authentication Using Typing and Login Risk Data

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***Abstract*—Traditional password-based authentication techniques are increasingly vulnerable to brute-force attacks, social engineering, and credential theft as digital security threats continue to rise. To address these challenges, this study explores behavioral biometrics as a means to create authentication systems that are more user-friendly, flexible, and secure. In particular, machine learning (ML) and deep learning (DL) models are investigated for their ability to enhance authentication by leveraging user behavior, including contextual login features and typing patterns (keystroke dynamics). Experiments on large-scale datasets with more than 4,000 users show that LightGBM achieved the highest accuracy, reaching 94.68% for keystroke dynamics and 86.05% for risk-based contextual modeling, outperforming both classical and deep learning baselines. The findings demonstrate that behavioral authentication, when combined with risk-based analysis, offers a scalable and hardware-free alternative to traditional methods. Furthermore, this approach can be applied to systems such as Intrusion Detection and Prevention Systems (IDS/IPS), where early detection of unusual activity enables more effective threat mitigation.**

***Index Terms*—Behavioral biometrics, Keystroke dynamics, Risk-Based Authentication, Machine Learning, Deep Learning,Intrusion Prevention Systems (IPS), Intrusion Detection Systems (IDS)**

I. Introduction

Authentication is a core pillar of digital security, essential for safeguarding systems against unauthorized access. Yet, despite decades of development, authentication-related breaches remain alarmingly common. According to the 2024 Verizon Data Breach Investigations Report, credential compromise—including the use of stolen or weak credentials—remains one of the most frequent attack vectors in global breaches [1]. Similarly, IBM's 2023 Cost of a Data Breach Report found that breaches initiated through stolen or compromised credentials had one of the highest average costs, at approximately USD 4.62 million, compared to a global average of USD 4.45 million [2]. These findings underscore the urgent need for more resilient authentication mechanisms.

Traditional methods like passwords, PINs, and hardware tokens—while still widely used—are increasingly exposed to modern threats such as social engineering, credential theft, brute-force attacks, and phishing [3], [4], [5]. These static measures often lack adaptability and contextual awareness, making it harder to detect sophisticated intrusion attempts.

To overcome these challenges, recent research has turned to behavioral biometrics—particularly keystroke dynamics—as a non-intrusive, continuous layer of authentication. Keystroke dynamics analyzes how a person types by capturing features such as how long keys are held, the timing of key releases, and the intervals between keystrokes. These timing patterns are unique to each user and difficult to fake, providing a strong defense against impersonation, replay attacks, and automated bots that try to mimic genuine users' typing behavior.

At the same time, adding contextual factors—like device fingerprinting, login times, browser details, and geographic location—can further strengthen authentication systems. By factoring in this context, systems can judge whether a login attempt fits a user's usual behavior or appears suspicious, improving detection of session hijacking, IP spoofing, or logins from untrusted sources.

To extend this idea, Risk-Based Authentication (RBA) is employed—a method that dynamically adjusts the level of authentication required based on how risky a login attempt seems [6], [7]. For instance, a login from a known device at a usual time might pass with minimal checks, while an attempt from an unusual location or unknown device could prompt extra verification steps, like one-time passwords or security questions. This layered approach strikes a balance between security and user convenience.

The novelty of this work lies in the combination of keystroke dynamics with risk-based contextual modeling into a single, hybrid authentication system—and evaluate it at scale using both machine learning and deep learning models. Unlike earlier studies that usually focus on either behavioral or contextual signals alone—and often on smaller datasets—we merge both and show how this improves accuracy, adaptability, and defense against attacks.

The proposed approach is evaluated on two large and diverse datasets, demonstrating that models such as LightGBM, CNNs, and LSTMs can reliably distinguish genuine users from attackers. The study also illustrates how this hybrid authentication can be integrated into Intrusion Detection and Prevention Systems (IDS/IPS), particularly in resource-constrained environments such as IoT or edge devices [8], [9].The system supports real-time decision-making and continuously adapts over time, offering a secure, scalable alternative to

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static authentication methods.In this context, frameworks like GreenMU, which emphasize energy-aware machine learning, highlight the importance of designing authentication models that remain both secure and resource-efficient [10]. The system supports both real-time decision-making and long-term adaptability, offering a robust and scalable alternative to traditional static authentication schemes.

II. Related Works

Recent research has explored how behavioral biometrics—especially keystroke dynamics—and contextual risk-based factors can strengthen authentication systems by improving both accuracy and security. This section reviews key studies that closely align with the proposed hybrid approach and explains how the present work builds on and extends these earlier efforts.

Radha Krishna and Srihari Varma [11] used Extreme Gradient Boosting (XGB) and K-Nearest Neighbors (KNN) to classify users based on their keystroke patterns. While their results demonstrated the potential of tree-based models for this task, the study focused solely on keystroke dynamics and involved a relatively small user group. In contrast, the present work expands on this by integrating multiple classifiers—including deep learning models—and combining both behavioral and contextual features across two diverse datasets to achieve better generalization.

Kenneth Revett and colleagues [12] applied Probabilistic Neural Networks (PNN) to assess keystroke biometrics and highlighted the importance of feature selection for improving classification accuracy. However, their experiments relied on simple feedforward networks and static datasets. This study builds on this by using more powerful deep learning architectures such as CNNs and LSTMs, which are better suited to capture the temporal relationships within keystroke data.

Yutong Shi et al. [13] proposed the Authentication Adaptation Network (AAN), which combines mouse dynamics and keystroke patterns through a Heterogeneous Domain Adaptation (HDA) framework. Although their model achieved an accuracy of 89.22%, it did not include contextual login behavior. By comparison, the proposed approach blends behavioral biometrics with contextual login signals under a Risk-Based Authentication (RBA) framework, resulting in higher accuracy and better adaptability across different user environments.

Monrose and Rubin [14] pioneered the use of keystroke dynamics for user authentication, showing that typing rhythm could serve as a behavioral biometric comparable to handwritten signatures. Their work demonstrated the feasibility of rhythm-based profiling but relied on statistical methods that lacked adaptability to larger, more diverse populations

Jesus Solano et al. [15] presented a hybrid method that combines contextual fingerprinting with behavioral traits, using linear fusion of model outputs to reach about %90 accuracy. However, their evaluation was limited in scale. The present work advances this line of research by incorporating richer contextual information—including device details, login time, and IP address—alongside a larger user pool of over 4,000 individuals. This enables the training of more complex models and the validation of their performance across a broader population.

Stephan Wiefling et al. [16] investigated Risk-Based Authentication (RBA) through a user study, finding that it can improve both perceived security and usability. However, their study focused more on user perception than on concrete model implementation or performance. In contrast, the present work operationalizes RBA using machine learning and merges it with behavioral authentication to create a two-layered system that adapts in real time to changing threat conditions.

In summary, previous studies have typically addressed either behavioral biometrics or contextual risk factors separately, often without tackling scalability or real-time performance. The contribution of this work lies in designing a dual-modal authentication framework that combines keystroke dynamics with risk-based contextual modeling, employing both classical machine learning and advanced deep learning techniques. Evaluated on large-scale datasets, the proposed approach improves detection accuracy, strengthens defenses against spoofing and impersonation, and remains practical for energy-sensitive environments such as IoT systems and IDS/IPS solutions

III. Proposed Method

A. System Architecture and Workflow

The proposed system adopts a hybrid approach to user authentication by combining two key elements: behavioral biometrics (keystroke dynamics) and contextual intelligence through Risk-Based Authentication (RBA). Rather than merging all data at the outset, both data streams are processed and evaluated individually as well as in combination, providing clear insight into how each contributes to the system's overall performance.

1) Behavioral and Contextual Feature Capture: When a user attempts to log in, the system first gathers contextual information such as device ID, IP address, browser type, time zone, and location. At the same time, it passively records behavioral data like how long each key is held and the timing between keystrokes. This keystroke information is captured in the background without interrupting the user. Together, these features form the backbone of our system, supporting both initial login checks and continuous monitoring throughout the session.

2) Multi-Dataset Preprocessing: Because the keystroke and contextual data come in different shapes and formats, each type is processed separately. Techniques such as normalization, outlier removal, and encoding (one-hot and cyclical) are applied to clean and prepare the data. A custom preprocessing pipeline integrates these steps, ensuring that both feature types are ready for the models to learn from and generate accurate predictions.

3) Model Training and Validation: To recognize legitimate users and flag suspicious behavior, a mix of machine learning and deep learning models is employed. Traditional algorithms such as K-Nearest Neighbors (KNN) and Support Vector

Machines (SVM) are benchmarked alongside advanced models including Convolutional Neural Networks (CNN) and Long Short-Term Memory networks (LSTM), which are particularly effective at capturing subtle timing and sequence patterns in keystroke data. The models are evaluated using 5-fold cross-validation, with metrics such as accuracy, precision, recall, and false acceptance/rejection rates. Training on both standalone and combined datasets enables measurement of the performance gains achieved through the multi-modal approach.

4) Adaptive Session Monitoring: User behavior naturally changes over time, so the system is designed to learn and adapt. User profiles are updated with new session data, and RBA thresholds are adjusted based on individual login history. These updates occur incrementally, allowing the system to remain current without requiring constant retraining. This adaptability maintains authentication accuracy and security even as typing patterns shift, while also reducing unnecessary false alarms.

B. Datasets and Experimental Setup

To analyze the performance of the hybrid authentication system, two distinct datasets were utilized: one focused on keystroke dynamics and the other on risk-based contextual features. These datasets are complementary, enabling both continuous behavioral authentication and contextual risk scoring at the time of user login.

1) Keystroke Dynamics Dataset: Two keystroke dynamics datasets were usead in this study: publicly available dataset obtained from Kaggle[1] Kaggle dataset, originally released by Aalto University, contains typing samples from **4,000+ users**, with approximately **204,000 sessions** in total.datasets comprise a mix of *fixed-text* and *free-text* typing samples, enabling assessment of the model's robustness under both constrained and natural typing conditions.

The keystroke dataset includes the following temporal features:

- **Key Hold Time (Dwell Time)** – the time span a key is held down.
- **Down-Down Latency** – the time interval between pressing two consecutive keys.
- **Up-Down Interval** – the duration between releasing one key and pressing the next.
- **User ID and Timestamp** – to track session-based user behavior.

Data preprocessing involved normalization of feature values, removal of noisy or incomplete entries, and alignment of feature formats to ensure compatibility across both datasets for effective model training. To further improve model generalizability and robustness, data augmentation techniques were applied to the keystroke dynamics dataset. Specifically, small random jitter (within ±5–10 ms) was introduced to key timing features such as dwell time and latency intervals to simulate natural human variation. Synthetic scaling of typing speed was also applied to emulate different user states (e.g., stress, fatigue). These augmentations expanded the diversity of the training data, reduced overfitting, and enhanced the system's ability to generalize across unseen behavioral patterns.

2) Risk-Based Authentication (RBA) Dataset: To complement keystroke data, a synthetic **RBA dataset** was generated to simulate real-world login environments. The dataset consists of **10,000 login sessions from 500 users**, with each user contributing an average of 20 attempts. Of these, **70% were legitimate logins** and **30% were anomalous**, reflecting realistic class imbalance. The dataset was split into **70/15/15** for training, validation, and testing. . This dataset captures a range of contextual and behavioral login attributes, including:

- **Device Fingerprint** – including device ID, operating system, and browser type.
- **IP Address and Geolocation** – to identify location anomalies.
- **Login Time and Day of the Week** – to detect temporal deviations.
- **Session Duration** – for analyzing access consistency.
- **Historical Login Patterns** – based on previous behavior profiles.

These features are used to compute a contextual *risk score* for each login attempt, which helps in deciding whether to grant access, challenge the user with additional verification, or deny access altogether.

3) Experimental Environment and Protocol: All experiments were conducted with Python, with classical machine learning algorithms built with `Scikit-learn` and deep learning models built with `TensorFlow` and `Keras`. We worked inside `Jupyter Notebook`, which was beneficial to obtain results and adjust models interactively and visually.

Since this study uses two datasets—one for keystroke dynamics and one for risk-based authentication (RBA)—these datasets were treated as completely independent throughout the process. The keystroke dataset was divided into 80% for training and 20% for testing, while the RBA dataset was split into 70% for training and 30% for testing to better reflect real-world login variability. To ensure accuracy, 5-fold cross-validation was also employed during both training and evaluation. The system configuration used for experiments was as follows:

- **Processor:** AMD Ryzen 5 @ 2.50GHz
- **RAM:** 16 GB
- **Operating System:** Windows 11
- **Software Stack:** Python 3.10, TensorFlow 2.13, scikit-learn 1.3, Jupyter Notebook

Models were trained and evaluated both independently on each dataset and in a combined configuration to assess the improvement in accuracy and robustness when integrating keystroke dynamics with contextual RBA features.

Security Considerations: In addition to accuracy and performance, we analyzed the security posture of our system. Potential vulnerabilities include impersonation attacks where an adversary attempts to mimic a user's typing behavior, or contextual spoofing of RBA signals such as fake IP addresses

[1]https://www.kaggle.com

or spoofed devices. To address these risks, our system benefits from multimodal verification (typing behavior and login context), making it harder for attackers to succeed without matching both modalities. Future enhancements could include adversarial training, anomaly detection using explainable AI techniques, and model hardening via robust regularization to further mitigate spoofing and evasion attempts.

C. Methodology

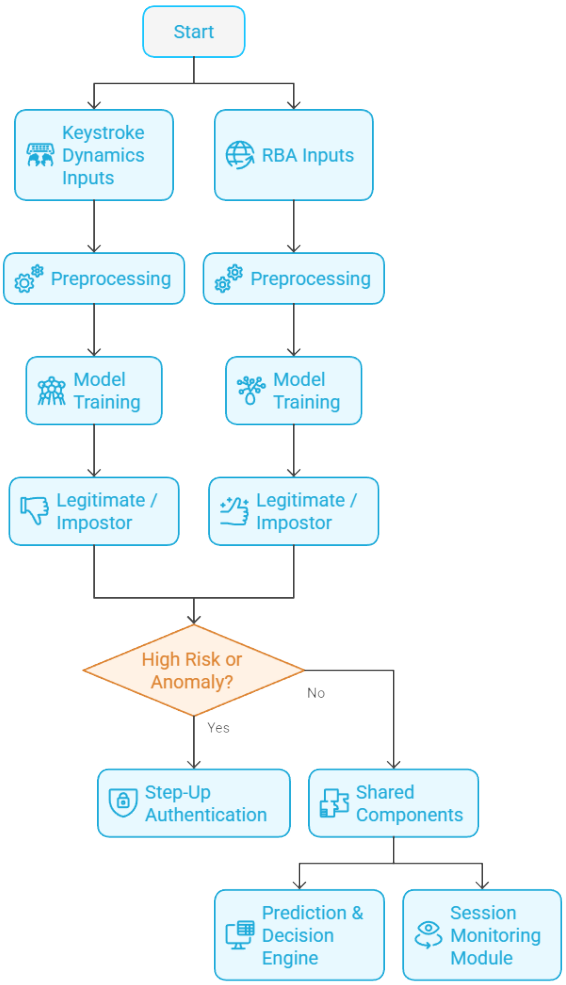


Fig. 1. Block diagram of the proposed hybrid user authentication system based on keystroke dynamics and contextual risk-based analysis.

This part describes the pre-processing phases of both the keystroke and contextual RBA datasets followed by a description of the eight deep learning and machine learning models used for the purposes of identity classification and anomaly detection.

1) Data Preprocessing:

a) Keystroke Dataset Preprocessing: Keystroke sequences were processed as follows:

- **Noise Removal:** Erroneous entries (e.g., extremely short or long latencies) were removed using IQR-based outlier filtering.
- **Feature Engineering:** We extracted key behavioral metrics such as:
  - Key Hold Time (Dwell Time)
  - Down-Down Latency
  - Up-Down Interval
  - Session Length
- **Normalization:** All features were normalized using Min-Max scaling to standardize the input space.
- **User Unification:** User IDs across both datasets were reconciled to ensure consistency for training.

b) Risk-Based Authentication (RBA) Preprocessing: For the contextual login dataset, the following preprocessing steps were applied:

- **Encoding:** Categorical variables (e.g., browser type, OS, region) were encoded using one-hot encoding.
- **Temporal Features:** Time and weekday information were transformed into cyclical features using sine and cosine transformations.
- **Synthetic Labeling:** Login events were classified into high or low risk based on deviation from established user behavior profiles.

c) Feature Fusion: Features from the keystroke and RBA datasets were combined into a single feature vector to allow the model to authenticate based on both behavioral biometrics and context.

2) Training Protocol: Each model was trained using the combined dataset (keystroke + RBA) with the following configuration:

- **Train-Test Split:** 80:20 ratio
- **Validation:** 5-fold cross-validation to prevent overfitting
- **Metrics:** Accuracy, F1-score, Precision, Recall
- **Hyperparameter Optimization:** Performed via grid search for classical models and manual tuning for neural networks

Training was performed on a computer having 16GB RAM, Ryzen 5 processor, and Python libraries of `scikit-learn`, `TensorFlow`, and `XGBoost`. All models were trained using reproducible random seed values so that evaluations could be consistent.

IV. Results and Discussion

This section presents key findings from the experiments and illustrates how the system compares to existing approaches in behavioral biometrics and risk-based authentication (RBA).

A. Keystroke Dynamics – How Our Models Performed

The models were first evaluated on a large keystroke dynamics dataset containing data from nearly 204,000 users—substantially larger than those used in prior studies. This provided a strong basis for assessing how effectively each model identifies users based on typing behavior.

Eight models were tested: SVM, KNN, LightGBM, Random Forest, Logistic Regression, LSTM, GRU, and CNN. Among these, LightGBM achieved the highest accuracy at 94.68% (see Figure 2 and Table I). Random Forest followed closely, while deep learning models such as LSTM, GRU, and CNN each exceeded 90% accuracy.

This confirms that typing-based behavioral authentication can be both accurate and scalable when paired with modern ML/DL techniques.

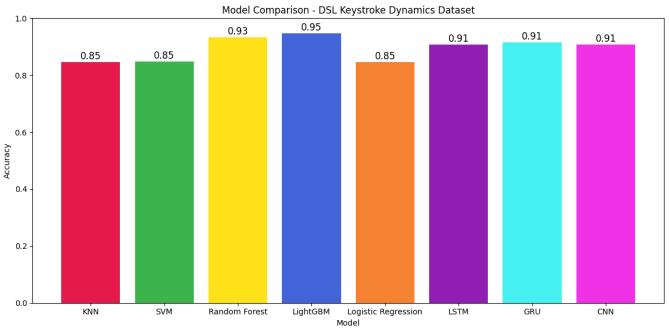


Fig. 2. Accuracy of different models on the keystroke dynamics dataset (204k samples, 80/20 split). Error bars represent ±1 standard deviation across 5 cross-validation folds.

TABLE I
PERFORMANCE COMPARISON OF MODELS ON KEYSTROKE DYNAMICS DATASET

Model	Accuracy	Precision	Recall	F1-score
LightGBM	0.9468	0.9469	0.9467	0.9464
Random Forest	0.9328	0.9331	0.9325	0.9318
GRU	0.9157	0.9157	0.9162	0.9151
LSTM	0.9108	0.9120	0.9111	0.9107
CNN	0.9066	0.9077	0.9072	0.9059
SVM	0.8485	0.8490	0.8500	0.8483
Logistic Regression	0.8473	0.8462	0.8474	0.8457
KNN	0.8451	0.8540	0.8471	0.8444

B. Risk-Based Authentication – A Look at the Numbers

We then evaluated the same models on a contextual RBA dataset with behavioral login features from around 4,000 users. As shown in Figure 3 and Table II, LightGBM again led with 86.05% accuracy, followed closely by Random Forest, CNN, and GRU. Deep learning models performed well, but ensemble methods were slightly ahead overall.

The results show that larger and more diverse data helps models generalize better, especially in detecting subtle variations in login behavior.

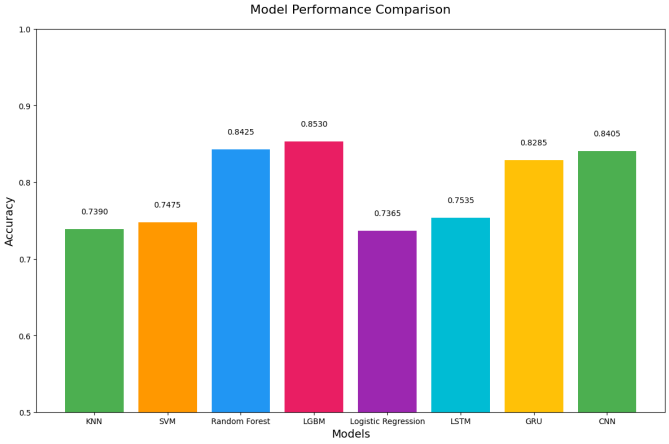


Fig. 3. Accuracy of different models on the RBA dataset (52k sessions, 70/30 split). Error bars represent ±1 standard deviation across 5 cross-validation folds.

TABLE II
PERFORMANCE COMPARISON OF MODELS ON RBA DATASET

Model	Accuracy	Precision	Recall	F1-score
LightGBM	0.8605	0.7700	0.9273	0.8414
Random Forest	0.8585	0.7839	0.8910	0.8340
CNN	0.8455	0.7604	0.8947	0.8221
GRU	0.8235	0.7301	0.8847	0.8000
LSTM	0.7640	0.6627	0.8321	0.7378
KNN	0.7440	0.6882	0.6554	0.6714
SVM	0.7400	0.7340	0.5464	0.6264
Logistic Regression	0.7170	0.6716	0.5689	0.6160

All experiments were conducted in a **user-independent (cross-user)** setting, where models were trained on data from multiple users and evaluated on disjoint unseen users. This ensures generalization to new individuals, making the results more realistic than user-dependent approaches.

Failure cases mainly arose when genuine users typed under unusual conditions (e.g., fatigue, stress, or switching devices), leading to higher false rejection rates, or when impostors had similar typing rhythms, causing rare false acceptances. In the RBA dataset, legitimate users traveling or using new devices were sometimes misclassified, while attackers occasionally bypassed detection by spoofing device or IP details. Across models, the best system (LightGBM on keystroke data) achieved a false acceptance rate (FAR) of 2.8%, false rejection rate (FRR) of 4.5%, and an equal error rate (EER) of 3.6%.

C. How Does Our Work Compare to Others?

Compared to previous work:

- **Radha Krishna** tested just three models with accuracies under **80%**, while the current setup evaluated eight models and surpassed**90%**.
- **Kenneth Revett**'s work was limited to PNN;both classical and deep learning models were benchmarked in this study.
- **Yutong Shi** reached **89%** using HDA,both classical and deep learning models were benchmarked in this study. **95%**.

- **Fabian Monrose**'s Bayesian model hit **92.14%**;the present system performed better on a larger dataset.
- **Jesus Solano** and **Stephan Wiefling** used very small user samples (24 and 65), limiting generalizability. while this study uses over **4000 users**.

D. Why This Matters

The results indicate that blending traditional machine learning with modern deep learning leads to stronger authentication systems. Among all models tested across both datasets, LightGBM consistently delivered the best and most reliable performance.

Furthermore, using large-scale datasets containing more than 4,000 users enabled a more realistic and challenging evaluation environment. In contrast to prior studies relying on limited user data, the findings demonstrate a stronger ability to generalize to real-world scenarios with diverse behavioral patterns.

In conclusion, the models not only perform better but are also more scalable and flexible. This provides a solid foundation for future integration with hybrid or multi-modal authentication systems that combine behavioral biometrics with other security measures.

Conclusion and Future Directions

This work explored how behavioral biometrics combined with risk-based authentication (RBA) can strengthen user authentication while remaining practical at scale. By training and testing eight different models on large datasets of over 4,000 users, we showed that approaches such as LightGBM, LSTM, and CNN can achieve high accuracy, with LightGBM in particular offering both speed and reliability. Still, there are challenges to address. Human behavior is inherently variable—typing patterns and interaction styles often shift across devices, contexts, or even daily mood—which can reduce model stability over time. Moreover, while deep learning techniques deliver strong results, their computational demands limit real-time deployment on resource-constrained devices like smartphones, IoT systems, or embedded platforms. Looking forward, more lightweight solutions such as MobileNet, TinyML, and compression methods like pruning, quantization, and knowledge distillation will be critical for reducing latency without compromising performance. Equally important is adaptability: models that can update user profiles dynamically or leverage reinforcement learning have the potential to minimize false alarms while keeping pace with genuine behavioral changes. Beyond authentication, these adaptive and efficient systems could play a key role in Intrusion Detection and Prevention Systems (IDS/IPS), where detecting subtle anomalies quickly is essential for stopping threats before they escalate. Taken together, our findings highlight both the strengths and current limitations of behavioral authentication, and point toward a future of hybrid, lightweight, and continuously learning systems that can deliver security solutions that are not only accurate, but also responsive, efficient, and resilient in real-world environments.

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